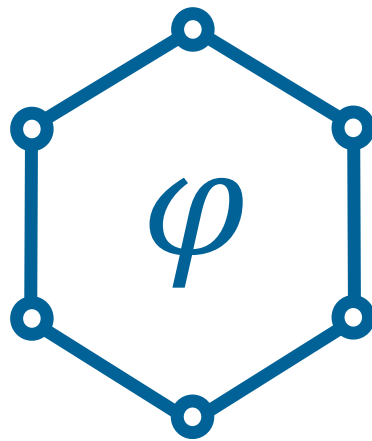


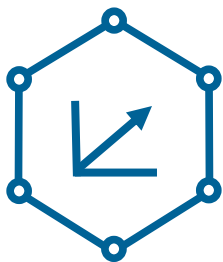


Logical Knowledge in KGs

Emanuel Sallinger

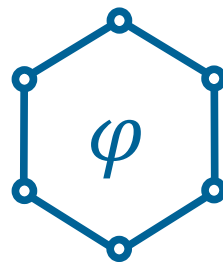


Representations



KG Embeddings

Widely-applied, large family of ML models



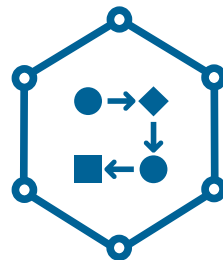
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



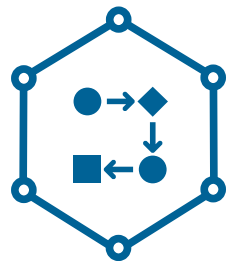
Graph Neural Networks

Using the KG structure as a neural network.



Data Models

Overview of data models in different communities



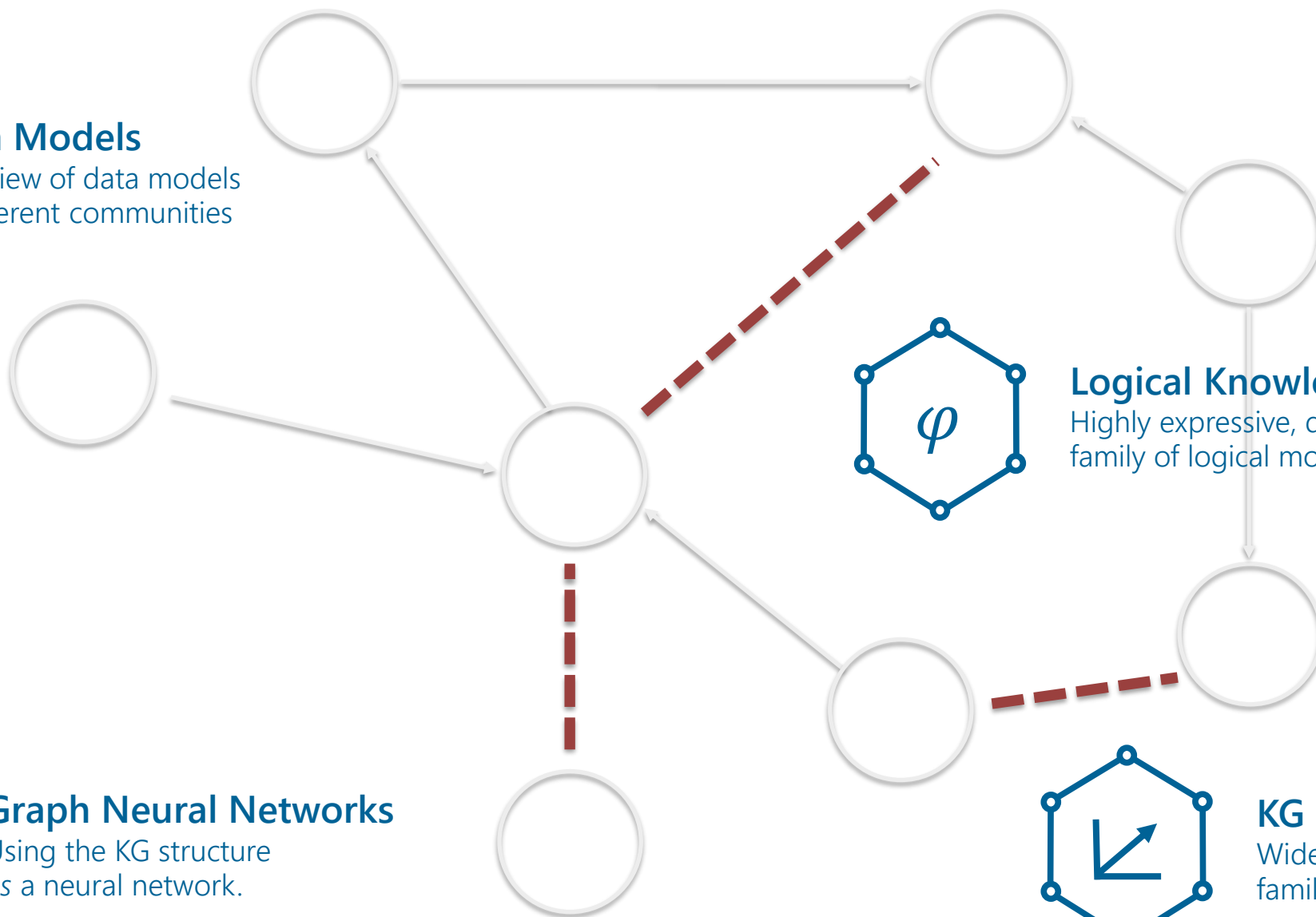
Data Models

Overview of data models in different communities



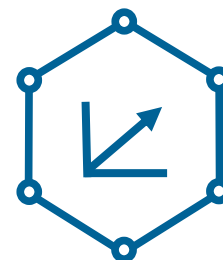
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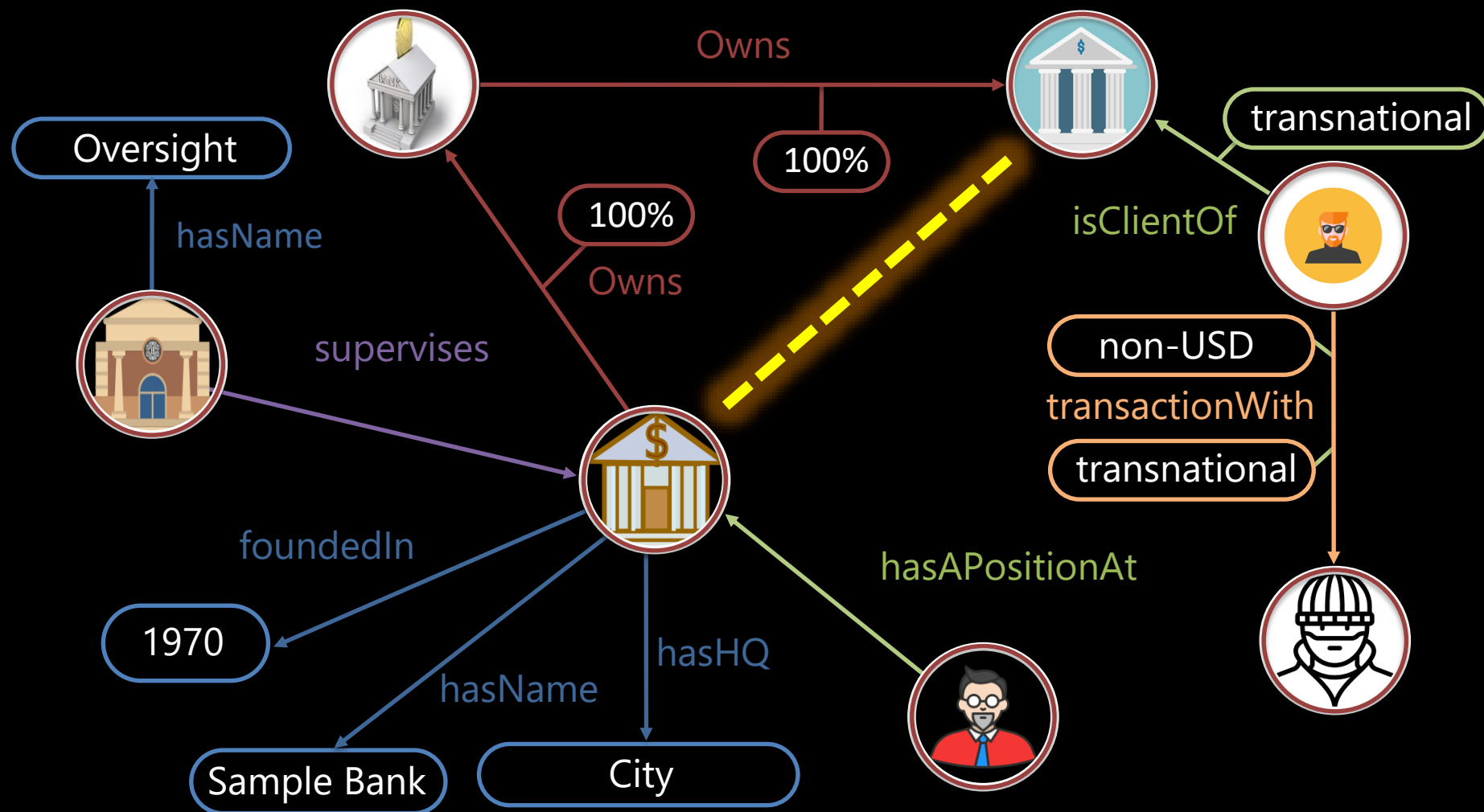
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Highly expressive, diverse family of logical models.



KG Embeddings

Widely-applied, large family of ML models

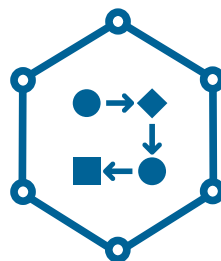




Representations



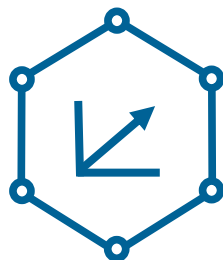
Graph



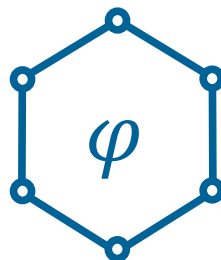
Knowledge Graph
Data Models



Knowledge



Knowledge Graph
Embeddings



Logical
Knowledge



Graph Neural
Networks

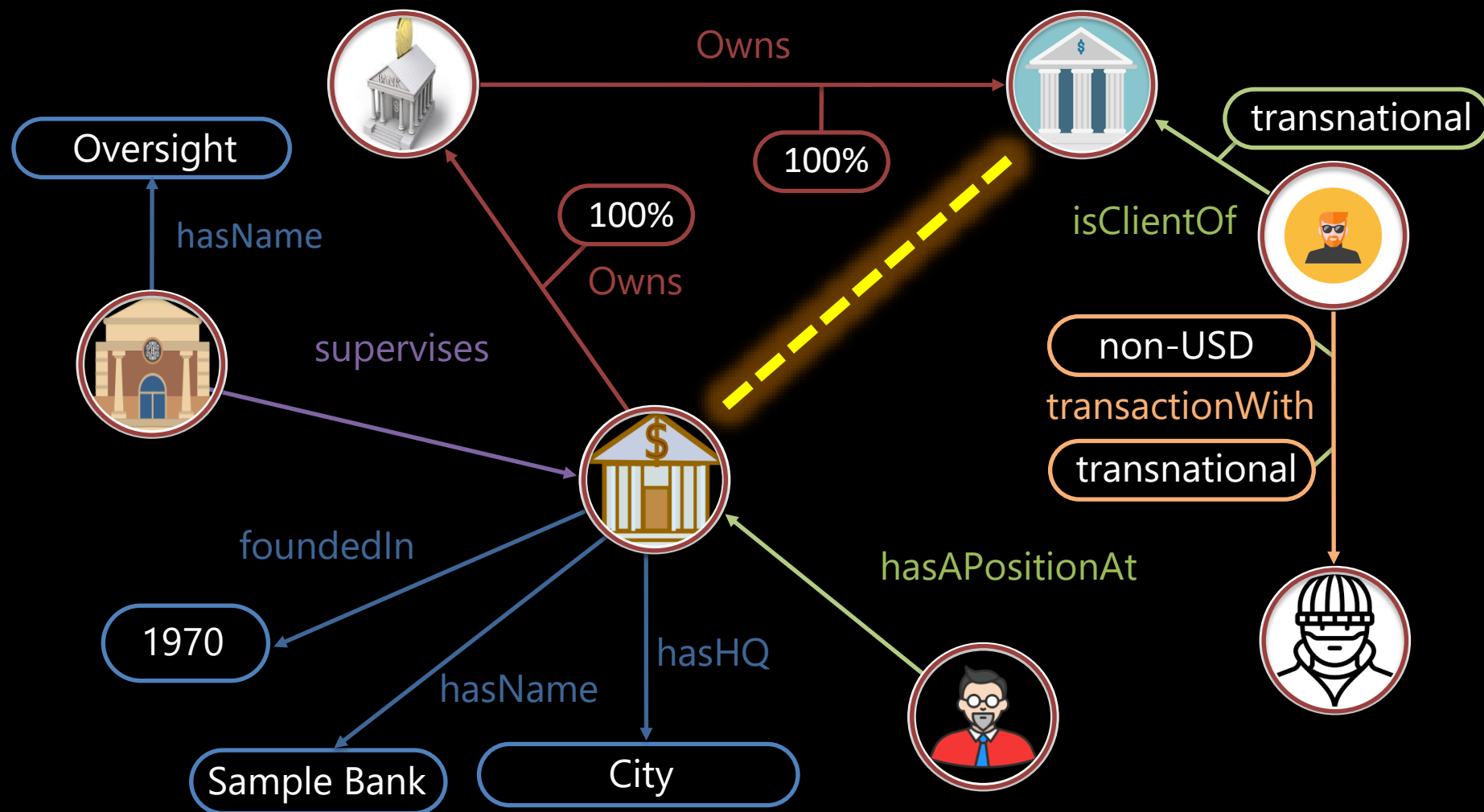
What is a Logic?

$$\models: \mathbb{F} \rightarrow \mathbb{F}$$

What is a Logic?

$$\models: \mathbb{F} \rightarrow \mathbb{F}$$

$\models$ is called a *logical consequence* operation
 $\mathbb{F}$ are sets of formulas



x **controls** y if
 x **directly holds** over 50% of y

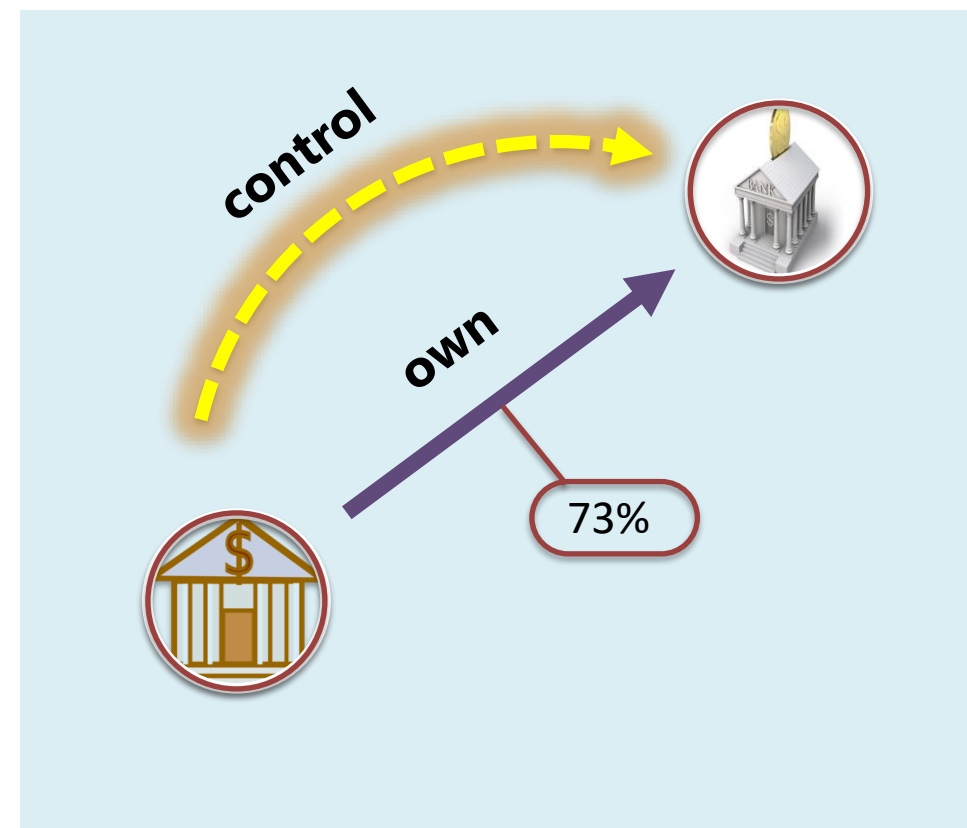
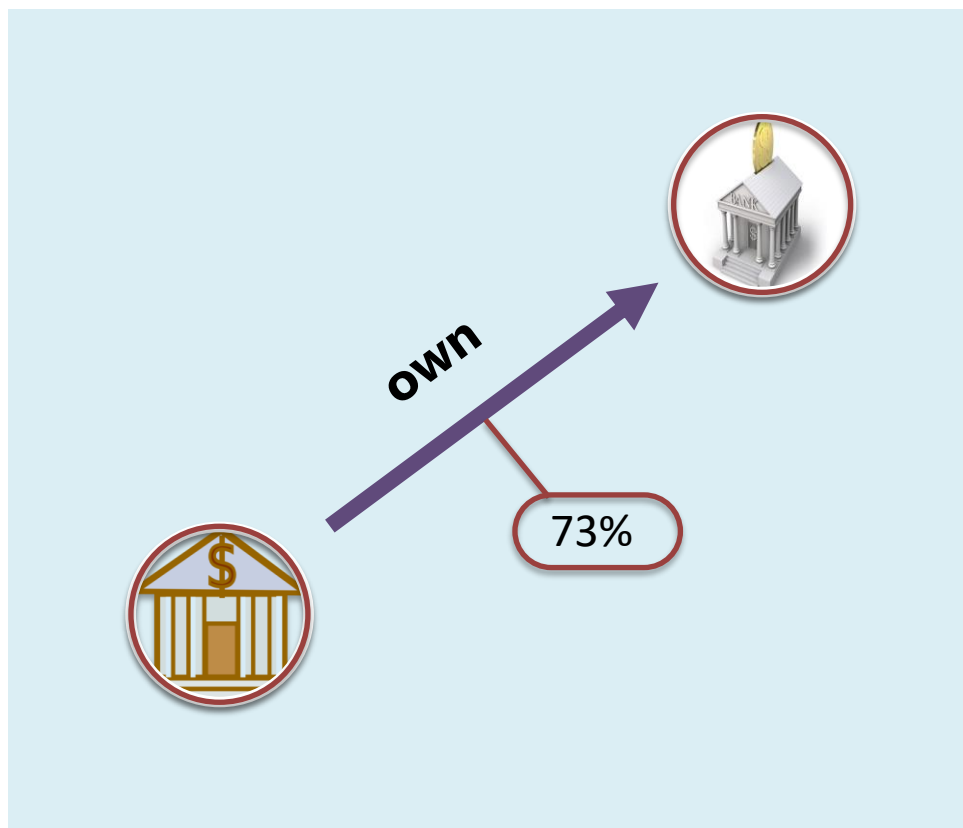


First-order Logic

$\forall x, y (\text{own}(x, y, w), w > 0.5 \rightarrow \text{control}(x, y))$



x **controls** y if
 x **directly holds** over 50% of y



x **controls** y if
 x **directly holds** over 50% of y



Datalog

```
control(X,Y) :- own(X,Y,W), W > 0.5.
```

x **controls** y if
 x **directly holds** over 50% of y



SQL

```
SELECT x,y INTO control
FROM company
WHERE w > 0.5
```

x **controls** y if
 x **directly holds** over 50% of y



Relational Calculus

$$control = \{(x, y) \mid own(x, y, w), w > 0.5 \}$$

x **controls** y if
 x **directly holds** over 50% of y



Relational Algebra

$control = \sigma_{w>0.5} own$

x **controls** y if
 x **directly holds** over 50% of y



Cypher (Graph DBs)

```
MATCH (x:Company) -[o:OWN]-> (y:Company)
WHERE o.w > 0.5
CREATE (x) -[:CONTROL]-> (y)
```



...

Cypher

SPARQL

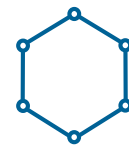
Datalog

SQL

First-order Logic

Relational Algebra

How does this help us?



Knowledge Graph

$$KG = (D, K; g, l, r, \dots)$$

Obtaining
Learning

Applying
Reasoning

Data and Knowledge

$$KG = (D, K; \dots)$$

Data: $\{own(c1, c2, 0.73)\}$

Knowledge: $\{\forall x, y (own(x, y, w), w > 0.5 \rightarrow control(x, y))\}$

Applying Knowledge

(Reasoning* with Knowledge)



Knowledge



Graph

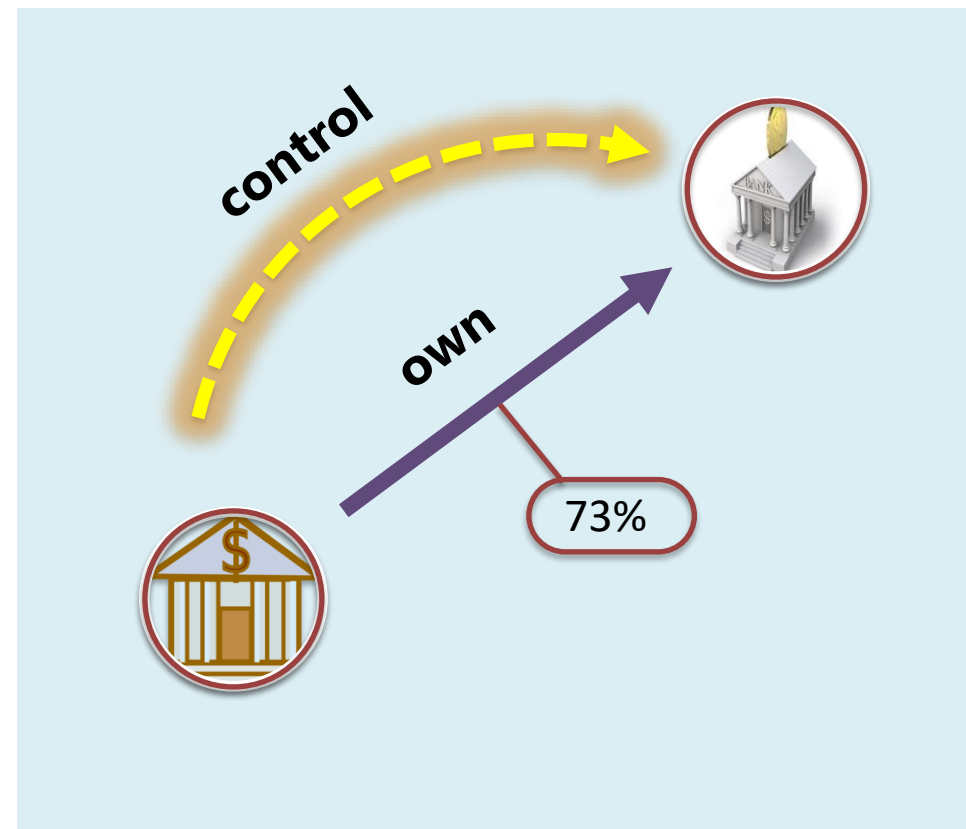
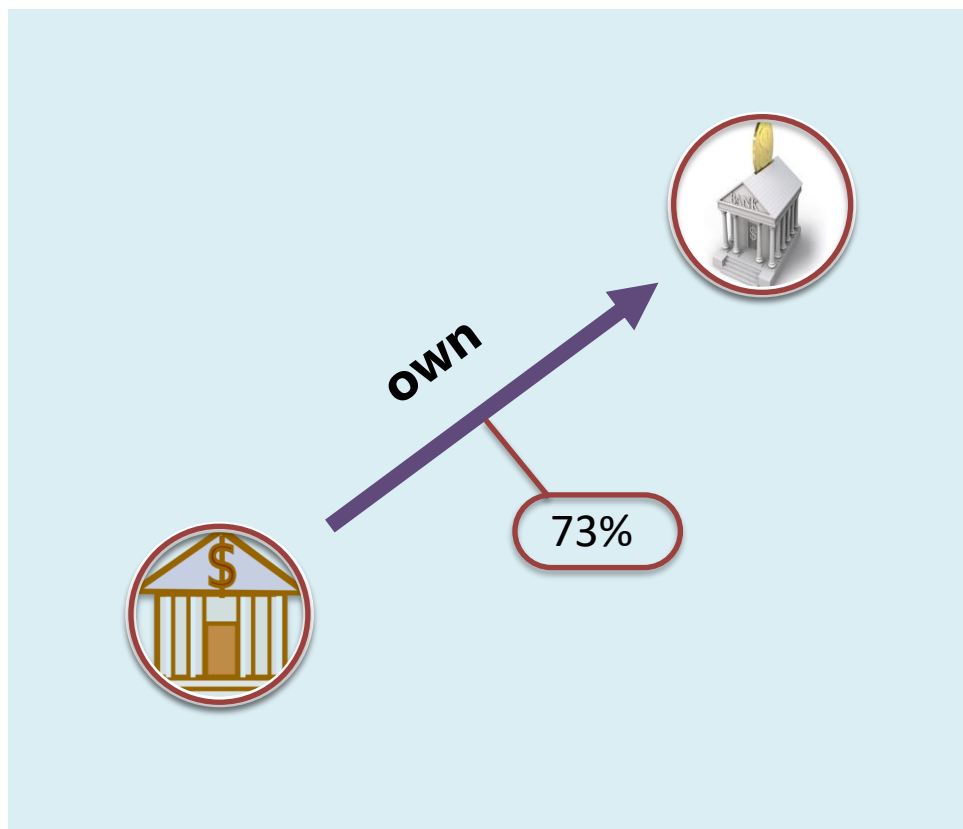


$$KG = (D, K; g, l, r)$$

$$r_K: \mathbb{D} \rightarrow \mathbb{D}$$



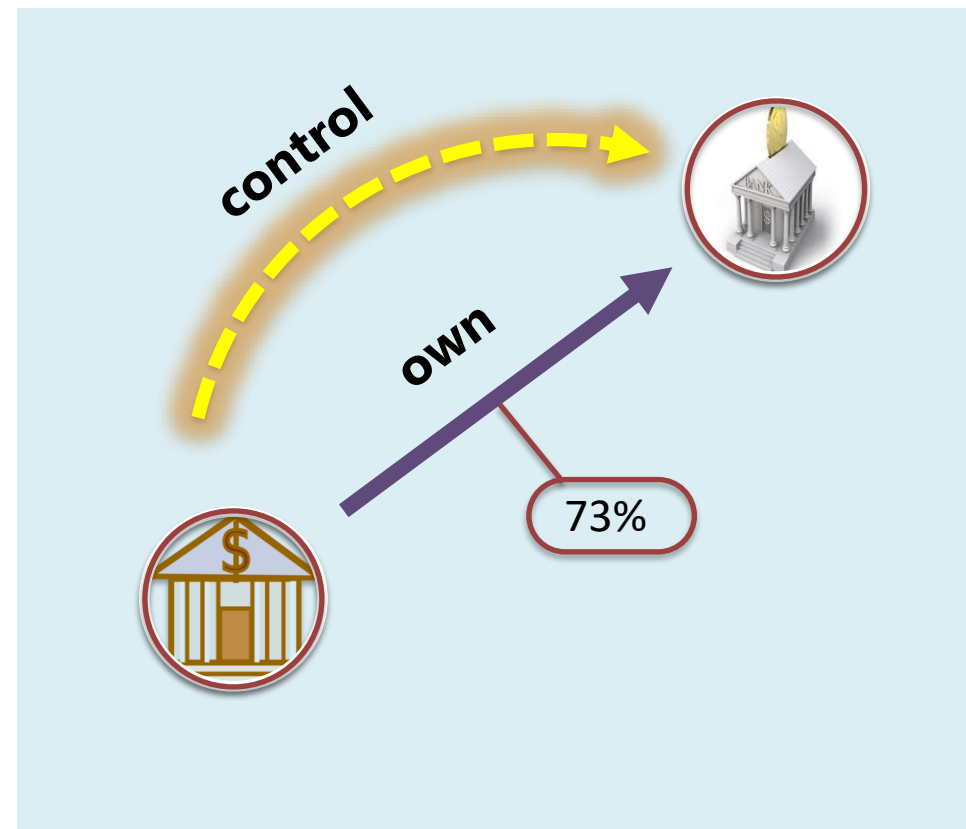
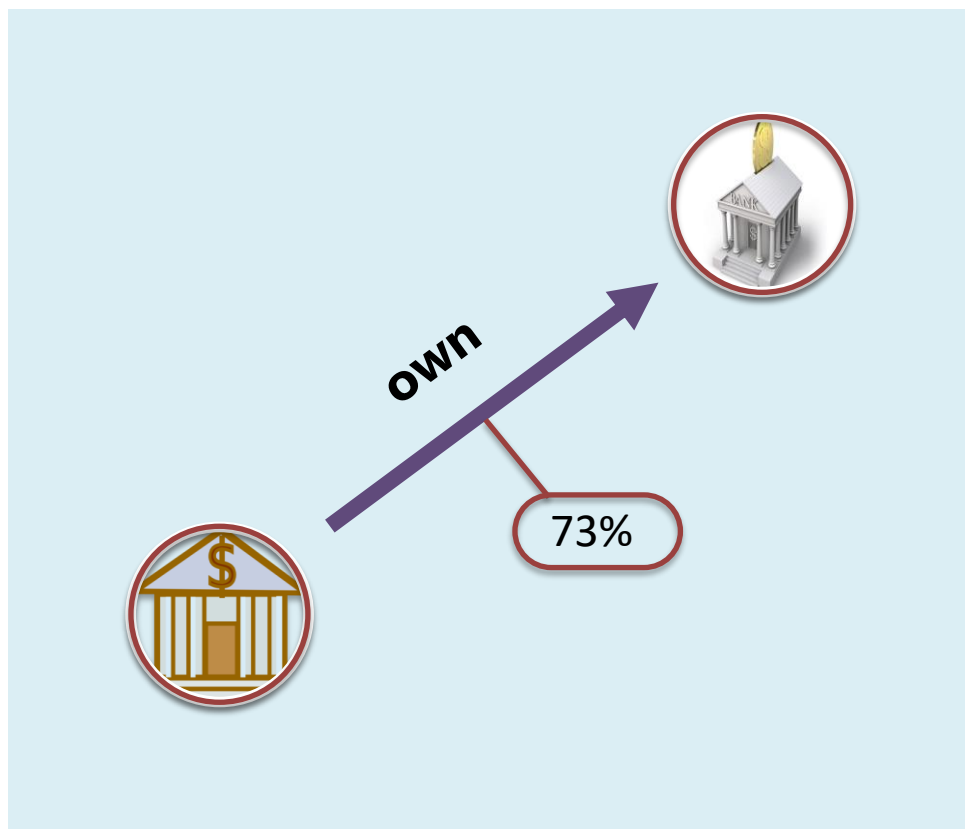
$$r_{\mathbb{K}}: \mathbb{D} \rightarrow \mathbb{D}$$





$$K = \{\forall x, y (own(x, y, w), w > 0.5 \rightarrow control(x, y))\}$$

$$r_K: \mathbb{D} \rightarrow \mathbb{D}$$



Logical Knowledge in KGs:

Reasoning with **Explicit** Knowledge

$$KG = (D, K; g, l, r)$$

$$r_{\mathbb{F}}: \mathbb{D} \rightarrow \mathbb{D}$$

Logical Knowledge in KGs:

Reasoning with **Explicit** Knowledge

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{F}$$

Logical Knowledge in KGs:

Reasoning with **Explicit** Knowledge

$$KG = (D, K; g, l, r)$$

$$r = \models$$

Logical Knowledge in KGs:

Learning **Explicit** Knowledge

$$KG = (D, K; g, l, r)$$

$$l: \mathbb{D} \rightarrow \mathbb{F}$$

Logical Knowledge in KGs:

Explicit Knowledge

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{F}$$

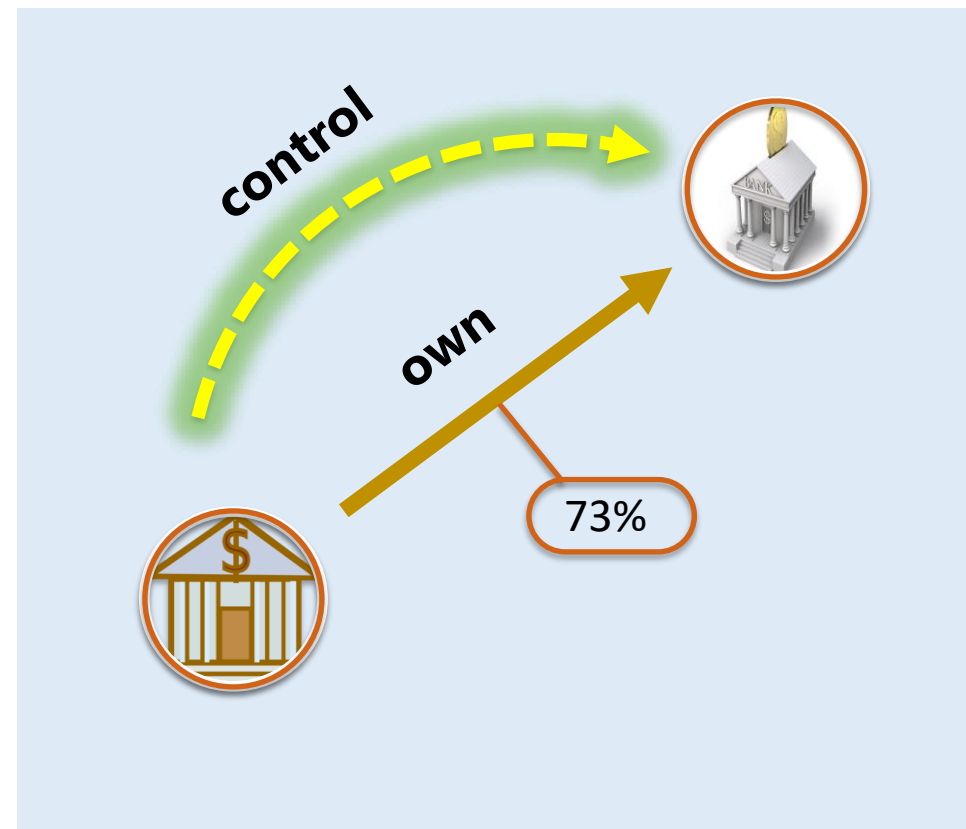
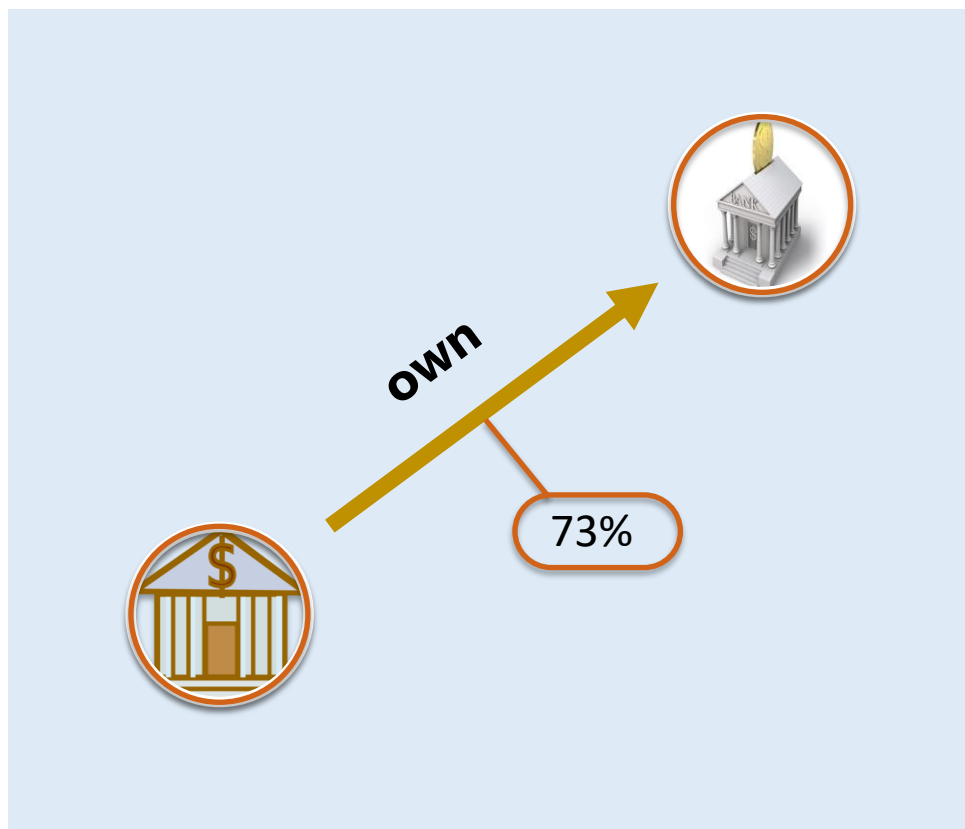
$$l: \mathbb{D} \rightarrow \mathbb{F}$$

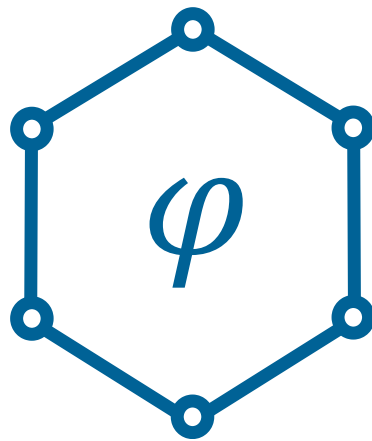
$$r = \models$$



$$K = \{\forall x, y (own(x, y, w), w > 0.5 \rightarrow control(x, y))\}$$

$$r_K: \mathbb{D} \rightarrow \mathbb{D}$$





Logical Knowledge in KGs

Emanuel Sallinger